

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [5]: transactions = pd.read_excel(r"D:\Forage\QVI_transaction_data.xlsx")
transactions
```

```
Out[5]:
```

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	P
--	------	-----------	----------------	--------	----------	-----------	---

	0	43390	1	1000	1	5	Natural Chip Compny SeaSalt175g
	1	43599	1	1307	348	66	CCs Nacho Cheese 175g
	2	43605	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g
	3	43329	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g
	4	43330	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g
	...	...	...	...	...	...	...
	264831	43533	272	272319	270088	89	Kettle Sweet Chilli And Sour Cream 175g
	264832	43325	272	272358	270154	74	Tostitos Splash Of Lime 175g
	264833	43410	272	272379	270187	51	Doritos Mexicana 170g
	264834	43461	272	272379	270188	42	Doritos Corn Chip Mexican Jalapeno 150g
	264835	43365	272	272380	270189	74	Tostitos Splash Of Lime 175g

264836 rows × 8 columns



```
In [7]: transactions.head(10)
```

Out[7]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PROD_C
0	43390	1	1000	1	5	Natural Chip Compny SeaSalt175g	
1	43599	1	1307	348	66	CCs Nacho Cheese 175g	
2	43605	1	1343	383	61	Smiths Crinkle Cut Chips Chicken 170g	
3	43329	2	2373	974	69	Smiths Chip Thinly S/Cream&Onion 175g	
4	43330	2	2426	1038	108	Kettle Tortilla ChpsHny&Jlpno Chili 150g	
5	43604	4	4074	2982	57	Old El Paso Salsa Dip Tomato Mild 300g	
6	43601	4	4149	3333	16	Smiths Crinkle Chips Salt & Vinegar 330g	
7	43601	4	4196	3539	24	Grain Waves Sweet Chilli 210g	
8	43332	5	5026	4525	42	Doritos Corn Chip Mexican Jalapeno 150g	
9	43330	7	7150	6900	52	Grain Waves Sour Cream&Chives 210G	



In [8]: `transactions.shape`

Out[8]: (264836, 8)

In [9]: `transactions.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   DATE                  264836 non-null int64
1   STORE_NBR             264836 non-null int64
2   LYLTY_CARD_NBR        264836 non-null int64
3   TXN_ID                264836 non-null int64
4   PROD_NBR              264836 non-null int64
5   PROD_NAME             264836 non-null object
6   PROD_QTY              264836 non-null int64
7   TOT_SALES             264836 non-null float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
```

```
In [10]: transactions.isnull().sum()
```

```
Out[10]: DATE                0
STORE_NBR                  0
LYLTY_CARD_NBR            0
TXN_ID                    0
PROD_NBR                  0
PROD_NAME                 0
PROD_QTY                  0
TOT_SALES                 0
dtype: int64
```

```
In [11]: transactions.duplicated().sum()
```

```
Out[11]: 1
```

```
In [ ]: #Removing duplicates
transactions=transactions.drop_duplicates()
```

```
In [ ]: #Checking for duplicates again
transactions.duplicated().sum()
```

```
Out[ ]: 0
```

```
In [16]: transactions['PROD_QTY'].describe()
```

```
Out[16]: count    264836.000000
mean         1.907309
std          0.643654
min          1.000000
25%          2.000000
50%          2.000000
75%          2.000000
max         200.000000
Name: PROD_QTY, dtype: float64
```

```
In [ ]: transactions = transactions[transactions['PROD_QTY'] < 20]
```

```
In [ ]: transactions['PROD_QTY'].describe()
```

```
Out[ ]: count    264833.000000
        mean      1.905812
        std       0.343437
        min       1.000000
        25%       2.000000
        50%       2.000000
        75%       2.000000
        max       5.000000
        Name: PROD_QTY, dtype: float64
```

PROD_QTY had a maximum value of 200 which is unrealistic for normal chip purchases.

Transactions with quantity ≥ 20 were treated as outliers and removed.

```
In [ ]: #Converting the date column to datetime format
transactions["DATE"] = pd.to_datetime(transactions["DATE"], unit="D", origin="18
```

```
In [24]: transactions["DATE"].head()
```

```
Out[24]: 0    2018-10-17
        1    2019-05-14
        2    2019-05-20
        3    2018-08-17
        4    2018-08-18
        Name: DATE, dtype: datetime64[ns]
```

```
In [ ]: ## Creating UNIT_PRICE to check pricing consistency across transactions
transactions['UNIT_PRICE'] = transactions['TOT_SALES'] / transactions['PROD_QTY']
```

```
In [26]: transactions['UNIT_PRICE'].describe()
```

```
Out[26]: count    264833.000000
        mean      3.824632
        std       1.109527
        min       1.320000
        25%       3.000000
        50%       3.800000
        75%       4.600000
        max       6.500000
        Name: UNIT_PRICE, dtype: float64
```

```
In [28]: ## Checking whether chip prices fall within a reasonable range
transactions['UNIT_PRICE'].unique()
```

```
Out[28]: array([3. , 2.1 , 1.45, 4.6 , 5.1 , 5.7 , 3.6 , 3.9 , 4.6 , 1.7 , 3.3 ,
        2.3 , 3.8 , 5.4 , 2.7 , 3.7 , 4.4 , 2.6 , 3.25, 2.9 , 3.1 , 4.2 ,
        4.3 , 1.5 , 1.9 , 5.9 , 4.4 , 2.8 , 1.8 , 1.9 , 6.5 , 2.4 , 3.8 ,
        3.3 , 5.4 , 2.7 , 5.1 , 3.7 , 1.85, 1.35, 5.9 , 1.44, 2.8 , 1.84,
        2.16, 1.65, 1.32, 2.2 , 2.3 , 1.48, 2.55, 2.04])
```

```
In [29]: transactions["PROD_NAME"].str.contains("salsa", case=False).sum()
```

```
Out[29]: 18094
```

UNIT_PRICE ranges from ~1.32 to 6.50 which is reasonable for chip products.

However some products contain "salsa" which are not chips and should be removed.


```
In [38]: transactions['PROD_NAME'].sample(10, random_state=42)
```

```
Out[38]: 81873          Kettle Honey Soy   Chicken 175g
217661    Infuzions Thai SweetChili PotatoMix 110g
224332          Tostitos Smoked      Chipotle 175g
241751          Pringles Chicken    Salt Crips 134g
178760          Smiths Crinkle      Original 330g
53873    Kettle Tortilla ChpsHny&Jlpno Chili 150g
14346          Tostitos Splash Of   Lime 175g
34587          Tostitos Lightly     Salted 175g
144505    Cobs Popd Sour Crm  &Chives Chips 110g
246625    Doritos Corn Chips  Cheese Supreme 170g
Name: PROD_NAME, dtype: object
```

```
In [39]: transactions = transactions[~transactions["PROD_NAME"].str.contains("salsa", cas
```

Removing salsa products since analysis focuses only on chips category

```
In [40]: transactions.shape
```

```
Out[40]: (246739, 9)
```

```
In [43]: # Extracting pack size (grams) from product name for size preference analysis
transactions["PACK_SIZE"] = transactions["PROD_NAME"].str.extract("(\\d+)").astype
```

```
In [ ]: # Extracting brand name from product name for brand preference analysis
transactions["BRAND"] = transactions["PROD_NAME"].str.split().str[0]
```

```
In [44]: transactions[["PROD_NAME", "BRAND", "PACK_SIZE"]].head(10)
```

```
Out[44]:
```

	PROD_NAME	BRAND	PACK_SIZE
0	Natural Chip Compny SeaSalt175g	Natural	175
1	CCs Nacho Cheese 175g	CCs	175
2	Smiths Crinkle Cut Chips Chicken 170g	Smiths	170
3	Smiths Chip Thinly S/Cream&Onion 175g	Smiths	175
4	Kettle Tortilla ChpsHny&Jlpno Chili 150g	Kettle	150
6	Smiths Crinkle Chips Salt & Vinegar 330g	Smiths	330
7	Grain Waves Sweet Chilli 210g	Grain	210
8	Doritos Corn Chip Mexican Jalapeno 150g	Doritos	150
9	Grain Waves Sour Cream&Chives 210G	Grain	210
10	Smiths Crinkle Chips Salt & Vinegar 330g	Smiths	330

```
In [ ]: # Loading the purchase behaviour data
customers = pd.read_csv("QVI_purchase_behaviour.csv")
customers.head()
```

```
Out[ ]:  

|   | LYLTY_CARD_NBR | LIFESTAGE              | PREMIUM_CUSTOMER |
|---|----------------|------------------------|------------------|
| 0 | 1000           | YOUNG SINGLES/COUPLES  | Premium          |
| 1 | 1002           | YOUNG SINGLES/COUPLES  | Mainstream       |
| 2 | 1003           | YOUNG FAMILIES         | Budget           |
| 3 | 1004           | OLDER SINGLES/COUPLES  | Mainstream       |
| 4 | 1005           | MIDAGE SINGLES/COUPLES | Mainstream       |


```

```
In [47]: customers.shape
```

```
Out[47]: (72637, 3)
```

```
In [48]: customers.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   LYLTY_CARD_NBR   72637 non-null  int64
1   LIFESTAGE        72637 non-null  object
2   PREMIUM_CUSTOMER 72637 non-null  object
dtypes: int64(1), object(2)
memory usage: 1.7+ MB
```

```
In [49]: customers.isnull().sum()
```

```
Out[49]: LYLTY_CARD_NBR    0
LIFESTAGE                0
PREMIUM_CUSTOMER        0
dtype: int64
```

```
In [50]: customers.duplicated().sum()
```

```
Out[50]: 0
```

```
In [51]: # Merging transaction dataset with customer dataset using Loyalty card number
data = pd.merge(transactions, customers, on="LYLTY_CARD_NBR", how="left")
```

```
In [52]: data.shape
```

```
Out[52]: (246739, 13)
```

```
In [55]: data.sample(5)
```

Out[55]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	PROD_NAME	PRC
168811	2019-01-01	194	194148	193888	4	Dorito Corn Chp Supreme 380g	
38742	2019-01-18	104	104099	104318	74	Tostitos Splash Of Lime 175g	
238435	2019-05-25	237	237030	240200	24	Grain Waves Sweet Chilli 210g	
210730	2019-02-07	213	213067	212284	98	NCC Sour Cream & Garden Chives 175g	
205410	2019-02-20	86	86207	85417	52	Grain Waves Sour Cream&Chives 210G	



In [56]:

```
# Calculating total chip sales by customer life stage and premium category
segment_sales = data.groupby(["LIFESTAGE", "PREMIUM_CUSTOMER"])[ "TOT_SALES" ].sum(
segment_sales.sort_values("TOT_SALES", ascending=False).head(10)
```

Out[56]:

	LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES
6	OLDER FAMILIES	Budget	156863.75
19	YOUNG SINGLES/COUPLES	Mainstream	147582.20
13	RETIREEES	Mainstream	145168.95
15	YOUNG FAMILIES	Budget	129717.95
9	OLDER SINGLES/COUPLES	Budget	127833.60
10	OLDER SINGLES/COUPLES	Mainstream	124648.50
11	OLDER SINGLES/COUPLES	Premium	123531.55
12	RETIREEES	Budget	105916.30
7	OLDER FAMILIES	Mainstream	96413.55
14	RETIREEES	Premium	91296.65

In [57]:

```
# Understanding which customer segments purchase chips most frequently
segment_transactions = data.groupby(["LIFESTAGE", "PREMIUM_CUSTOMER"])[ "TXN_ID" ].
segment_transactions = segment_transactions.sort_values("TXN_ID", ascending=False)
segment_transactions.head(10)
```

Out[57]:

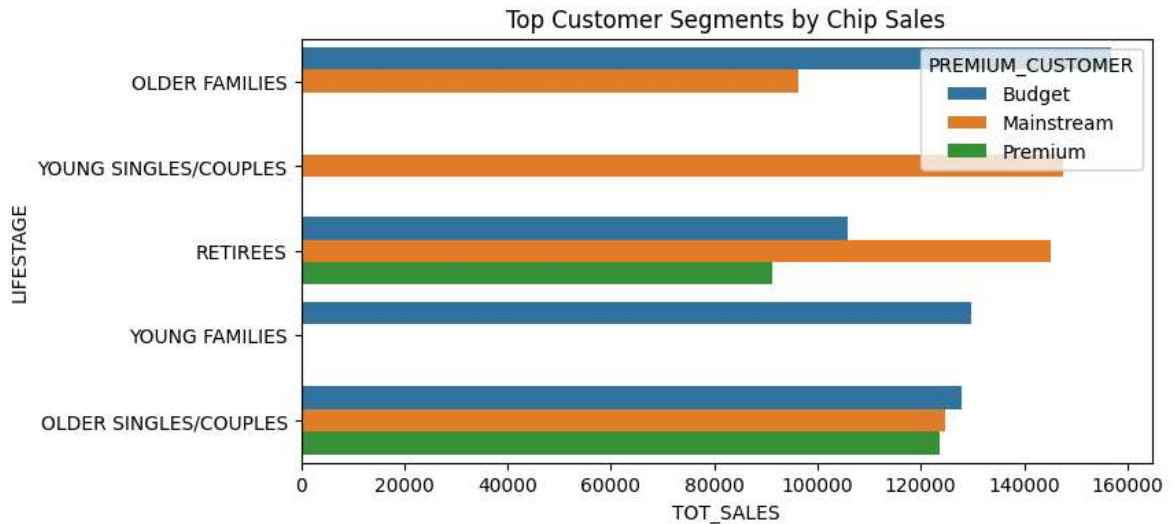
	LIFESTAGE	PREMIUM_CUSTOMER	TXN_ID
6	OLDER FAMILIES	Budget	21514
13	RETIREEES	Mainstream	19970
19	YOUNG SINGLES/COUPLES	Mainstream	19544
15	YOUNG FAMILIES	Budget	17763
9	OLDER SINGLES/COUPLES	Budget	17172
10	OLDER SINGLES/COUPLES	Mainstream	17061
11	OLDER SINGLES/COUPLES	Premium	16559
12	RETIREEES	Budget	14225
7	OLDER FAMILIES	Mainstream	13241
14	RETIREEES	Premium	12236

```
In [58]: # Analyzing preferred pack size across customer segments
pack_pref = data.groupby(["LIFESTAGE", "PREMIUM_CUSTOMER"])[ "PACK_SIZE" ].mean().r
pack_pref.sort_values("PACK_SIZE", ascending=False).head(10)
```

Out[58]:

	LIFESTAGE	PREMIUM_CUSTOMER	PACK_SIZE
19	YOUNG SINGLES/COUPLES	Mainstream	178.344249
1	MIDAGE SINGLES/COUPLES	Mainstream	177.898693
11	OLDER SINGLES/COUPLES	Premium	176.485657
12	RETIREEES	Budget	176.395641
14	RETIREEES	Premium	176.368421
4	NEW FAMILIES	Mainstream	175.629748
6	OLDER FAMILIES	Budget	175.546342
15	YOUNG FAMILIES	Budget	175.459720
9	OLDER SINGLES/COUPLES	Budget	175.334673
5	NEW FAMILIES	Premium	175.245296

```
In [61]: plt.figure(figsize=(8,4))
sns.barplot(data=segment_sales.sort_values("TOT_SALES", ascending=False).head(10),
            x="TOT_SALES", y="LIFESTAGE", hue="PREMIUM_CUSTOMER")
plt.title("Top Customer Segments by Chip Sales")
plt.show()
```



1.Highest Sales Segment

OLDER FAMILIES (Budget) generate the highest total chip sales among all customer segments.

2.Most Frequent Buyers

OLDER FAMILIES (Budget) and RETIREES (Mainstream) show the highest purchase frequency.

3.Strong Young Segment

YOUNG SINGLES/COUPLES (Mainstream) also contribute significantly to total sales and transaction count.

4.Pack Size Preference

Most customer segments prefer pack sizes around 175–180g.

Recommendation:

The supermarket should prioritize OLDER FAMILIES (Budget) and YOUNG SINGLES/COUPLES (Mainstream) as key customer segments for chip promotions.

These segments contribute the highest sales and transaction volume.

Promotions around 175g–180g pack sizes may perform well as this size is consistently preferred across segments.

```
In [62]: customer_counts = data.groupby(["LIFESTAGE", "PREMIUM_CUSTOMER"])["LYLTY_CARD_NBR"]
segment_sales = data.groupby(["LIFESTAGE", "PREMIUM_CUSTOMER"])["TOT_SALES"].sum()
sales_per_customer = segment_sales.merge(customer_counts, on=["LIFESTAGE", "PREMIUM_CUSTOMER"])
sales_per_customer["SALES_PER_CUSTOMER"] = sales_per_customer["TOT_SALES"] / sales_per_customer["LYLTY_CARD_NBR"]
```

```
sales_per_customer.sort_values("SALES_PER_CUSTOMER", ascending=False).head(10)
```

Out[62]:

	LIFESTAGE	PREMIUM_CUSTOMER	TOT_SALES	CUSTOMERS	SALES_PER_CUST
7	OLDER FAMILIES	Mainstream	96413.55	2788	34.5
6	OLDER FAMILIES	Budget	156863.75	4611	34.0
8	OLDER FAMILIES	Premium	75242.60	2231	33.7
15	YOUNG FAMILIES	Budget	129717.95	3953	32.8
17	YOUNG FAMILIES	Premium	78571.70	2398	32.7
16	YOUNG FAMILIES	Mainstream	86338.25	2685	32.1
11	OLDER SINGLES/COUPLES	Premium	123531.55	4682	26.3
9	OLDER SINGLES/COUPLES	Budget	127833.60	4849	26.3
1	MIDAGE SINGLES/COUPLES	Mainstream	84734.25	3298	25.6
10	OLDER SINGLES/COUPLES	Mainstream	124648.50	4858	25.6

Sales per customer analysis indicates that Older Families spend the most on average per customer across all premium categories. This suggests that this segment not only contributes large total sales but also demonstrates strong individual purchasing power in the chip category.

```
In [63]: txn_per_customer = data.groupby(["LIFESTAGE", "PREMIUM_CUSTOMER"]).agg(  
    TOTAL_TXN=("TXN_ID", "count"),  
    UNIQUE_CUSTOMERS=("LYLTY_CARD_NBR", "nunique")  
).reset_index()  
  
txn_per_customer["TXN_PER_CUSTOMER"] = txn_per_customer["TOTAL_TXN"] / txn_per_c  
  
txn_per_customer.sort_values("TXN_PER_CUSTOMER", ascending=False).head(10)
```

Out[63]:

	LIFESTAGE	PREMIUM_CUSTOMER	TOTAL_TXN	UNIQUE_CUSTOMERS	TXN_PI
7	OLDER FAMILIES	Mainstream	13241	2788	
6	OLDER FAMILIES	Budget	21514	4611	
8	OLDER FAMILIES	Premium	10403	2231	
17	YOUNG FAMILIES	Premium	10784	2398	
15	YOUNG FAMILIES	Budget	17763	3953	
16	YOUNG FAMILIES	Mainstream	11947	2685	
9	OLDER SINGLES/COUPLES	Budget	17172	4849	
11	OLDER SINGLES/COUPLES	Premium	16559	4682	
10	OLDER SINGLES/COUPLES	Mainstream	17061	4858	
1	MIDAGE SINGLES/COUPLES	Mainstream	11095	3298	



Transaction frequency analysis shows that Older Families and Young Families purchase chips more frequently than other customer segments. This suggests that these groups have a stronger habitual consumption pattern and represent a reliable base of repeat customers.

In [64]:

```
brand_sales = data.groupby("BRAND")["TOT_SALES"].sum().reset_index()

brand_sales = brand_sales.sort_values("TOT_SALES", ascending=False).head(10)
brand_sales
```

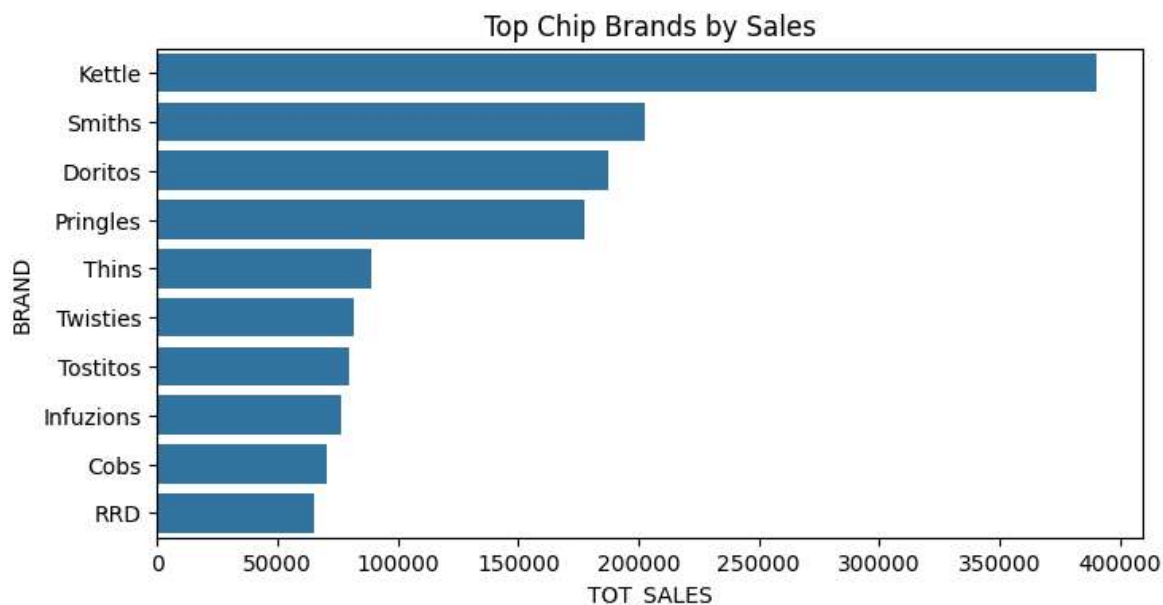
Out[64]:

	BRAND	TOT_SALES
12	Kettle	390239.8
19	Smiths	202902.6
6	Doritos	187277.9
15	Pringles	177655.5
22	Thins	88852.5
24	Twisties	81522.1
23	Tostitos	79789.6
10	Infuzions	76247.6
4	Cobs	70569.8
16	RRD	64954.5

In [66]:

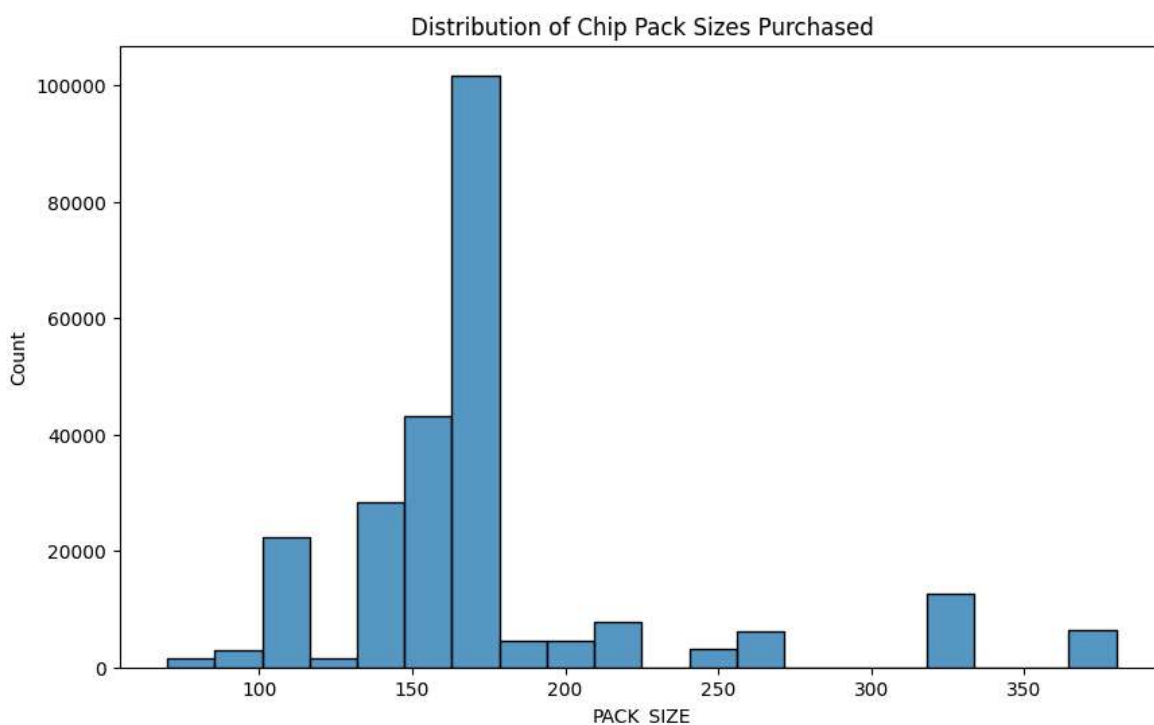
```
plt.figure(figsize=(8,4))
sns.barplot(data=brand_sales, x="TOT_SALES", y="BRAND")
```

```
plt.title("Top Chip Brands by Sales")  
plt.show()
```



The chip category is strongly driven by a few dominant brands. Kettle leads the category with the highest sales, followed by Smiths, Doritos, and Pringles. These brands represent the core of category demand and likely have strong customer loyalty and shelf presence. Focusing promotional strategies on these leading brands may have the greatest impact on category growth.

```
In [67]: plt.figure(figsize=(10,6))  
sns.histplot(data["PACK_SIZE"], bins=20)  
plt.title("Distribution of Chip Pack Sizes Purchased")  
plt.show()
```

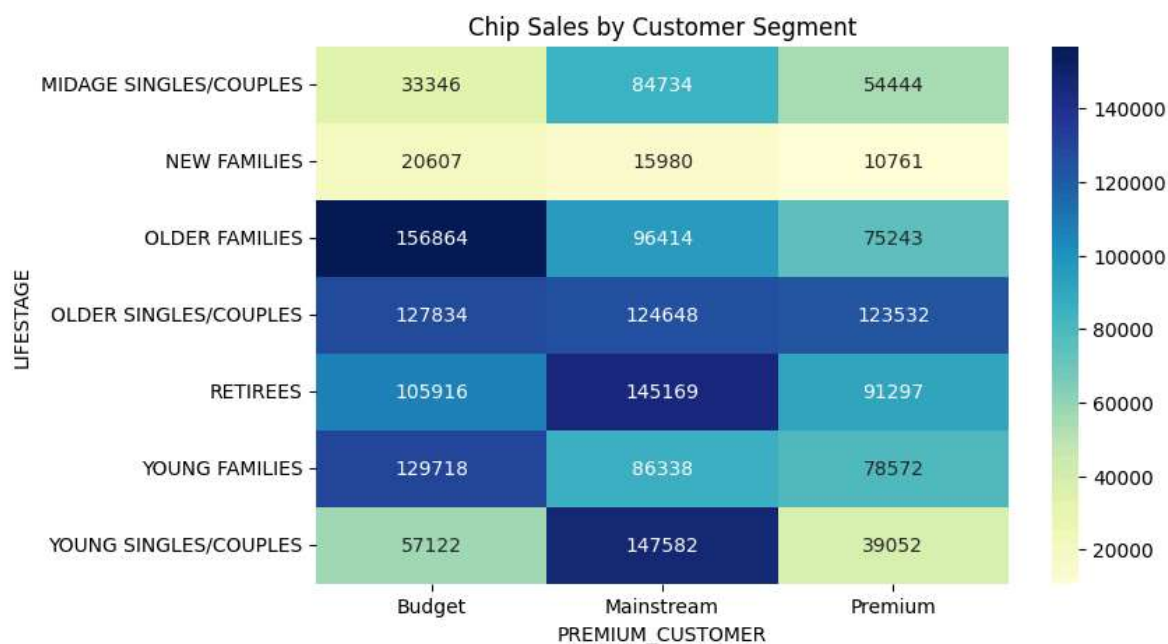


The distribution of chip pack sizes shows that most purchases cluster around the 150g–180g range. This suggests that standard pack sizes dominate consumer preference. Larger pack sizes such as 330g and

above appear less frequently, indicating that customers may prefer moderately sized packs for regular consumption.

```
In [69]: segment_matrix = data.pivot_table(
    values="TOT_SALES",
    index="LIFESTAGE",
    columns="PREMIUM_CUSTOMER",
    aggfunc="sum"
)

plt.figure(figsize=(8,5))
sns.heatmap(segment_matrix, annot=True, fmt=".0f", cmap="YlGnBu")
plt.title("Chip Sales by Customer Segment")
plt.show()
```

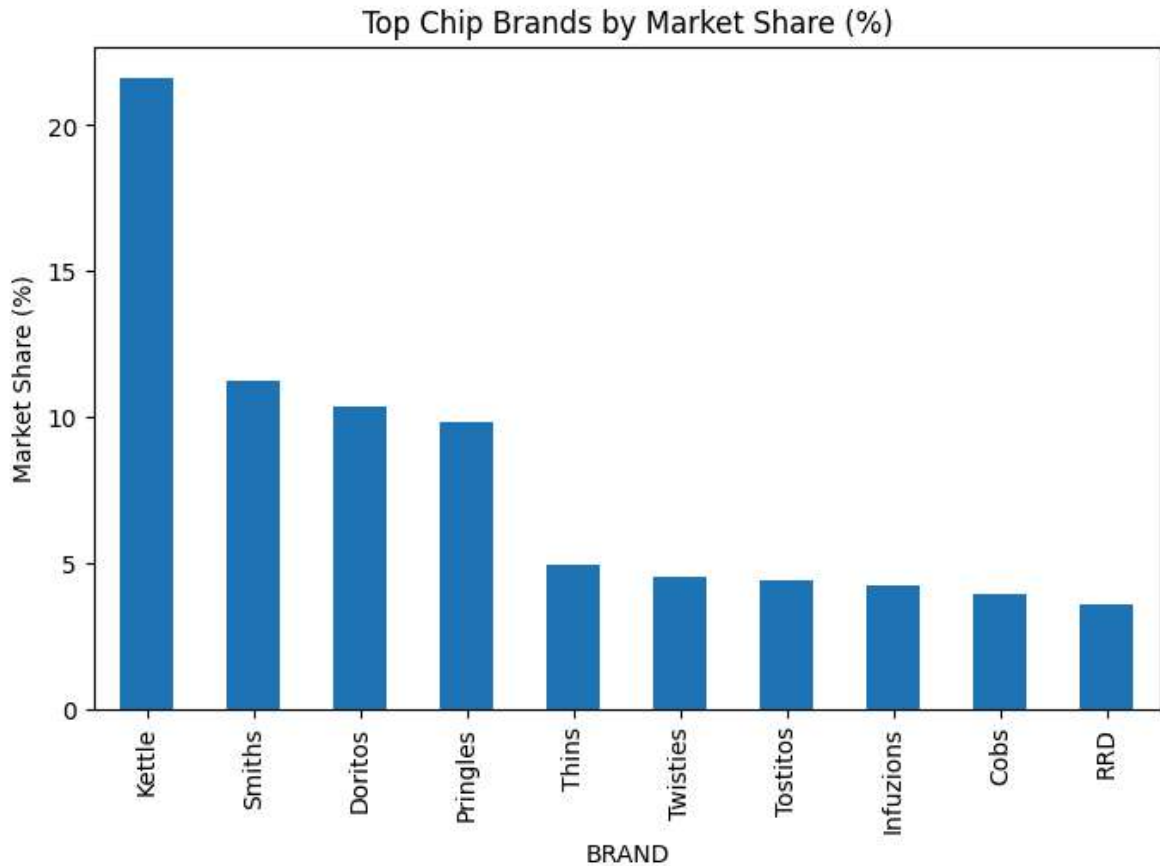


The heatmap highlights strong sales contributions from several key segments. Older Families (Budget) generate the highest total sales within the chip category, while Young Singles/Couples (Mainstream) and Retirees (Mainstream) also contribute significantly. These segments represent important customer groups for targeted marketing strategies.

```
In [70]: brand_share = data.groupby("BRAND")["TOT_SALES"].sum()
brand_share = (brand_share / brand_share.sum()) * 100
brand_share = brand_share.sort_values(ascending=False).head(10)
brand_share
```

```
Out[70]: BRAND
Kettle      21.617877
Smiths      11.240072
Doritos     10.374520
Pringles     9.841474
Thins        4.922108
Twisties     4.516030
Tostitos     4.420056
Infuzions    4.223842
Cobs         3.909312
RRD          3.598245
Name: TOT_SALES, dtype: float64
```

```
In [72]: plt.figure(figsize=(8,5))
brand_share.plot(kind="bar")
plt.title("Top Chip Brands by Market Share (%)")
plt.ylabel("Market Share (%)")
plt.show()
```



Brand Market Share Analysis

Insight

The market share analysis shows that the chip category is dominated by a few key brands. Kettle leads the category with approximately 21.6% of total chip sales, followed by Smiths (11.2%), Doritos (10.4%), and Pringles (9.8%).

These four brands together account for a large portion of category revenue, indicating that the chip market is relatively concentrated among a few leading brands.

Business Impact

Understanding brand market share helps category managers identify which brands have the strongest influence on overall category performance.

If this analysis were not performed, we would only observe absolute sales values without understanding how dominant certain brands are within the market. Market share analysis highlights the competitive positioning of brands and helps guide decisions around promotions, shelf space allocation, and brand partnerships.

Strategic Recommendation

Based on the analysis, the chip category is primarily driven by Older Families and Young Families across multiple spending and transaction metrics. These segments demonstrate both high total sales and strong purchase frequency, making them key drivers of category performance.

Additionally, Young Singles/Couples in the Mainstream segment contribute significantly to overall revenue and represent an important demographic for growth.

From a product perspective, standard pack sizes around 150g–180g dominate consumer purchasing behavior, suggesting that promotions and shelf placement strategies should prioritize these sizes.

Finally, leading brands such as Kettle, Smiths, Doritos, and Pringles account for a large portion of category sales, indicating that collaboration with these brands for promotional campaigns could further strengthen category performance.

Conclusion

This analysis identified the key customer segments, product characteristics, and brand drivers influencing chip purchases. By focusing marketing and promotional strategies on high-performing customer segments and popular pack sizes, the supermarket can better position the chip category for growth in the upcoming review period.